Homework 2 Pseudocode

Bayou City Family Health Patient Registration Form

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- Class: MIS 7375 Transaction Processing Systems I
- Date: October 23, 2026
- Version: 1.4

Main Program

MAIN

DO BEGINNING
DO THE MIDDLE
DO THE END
END MAIN

DO BEGINNING

DISPLAY THE HTML FORM (homework2.html)
SHOW TODAYS DATE IN THE BANNER
LOAD THE STATE LIST FROM homework2-states.js INTO THE STATE BOX
    PUT "Select" FIRST WITH NO VALUE SO THE USER HAS TO PICK ONE
SHOW THE SLIDER VALUE NEXT TO THE BAR
SET-CONDITIONS (onblur, onchange, oninput, onclick, onsubmit and onreset)
END BEGINNING

DO MIDDLE (javascript, homework2.js)

IF THE USER ENTERED AND THEN LEFT A FIELD
    CALL SHOW-RESULT FOR THAT FIELD
ENDIF
IF THE USER IS TYPING IN A PASSWORD BOX
    CALL SHOW-RESULT FOR THAT PASSWORD BOX
ENDIF
IF THE USER MOVED THE SLIDER
    CALL SHOW-SLIDER-VALUE
ENDIF
IF THE REVIEW BUTTON WAS CHOSEN
    CALL REVIEW-FORM
ENDIF

```
IF THE SUBMIT BUTTON WAS CHOSEN
        LET THE BROWSER CHECK REQUIRED FIELDS AND PATTERNS
        IF THE BROWSER CHECKS PASS
                CALL REVIEW-FORM
                IF NO ERROR CONDITIONS
                        DISPLAY THANK YOU PAGE
                ELSE
                        SCROLL TO THE REVIEW AREA
                        DISPLAY "FIX YOUR ERRORS" MESSAGE
                        DO NOT SEND THE FORM
                ENDIF
        ENDIF
ENDIF
IF THE RESET BUTTON WAS CHOSEN
        CLEAR THE FORM
        CLEAR THE REVIEW AREA AND EVERY ERROR MESSAGE
        PUT THE SLIDER VALUE BACK TO 5
ENDIF
END MIDDLE
```

THE-END

```
SHOW THE THANK YOU PAGE (thankyou.html)
GIVE A LINK BACK TO THE FORM
END THE-END
```

Helper Modules

GET-FIELD

```
FIND THE FIELD IN THE FORM BY ITS NAME (NOT ITS ID)
RETURN THE FIELD
END GET-FIELD
```

GET-VALUE

```
CALL GET-FIELD
TRIM THE EXTRA SPACES OFF THE ENDS
PUT THE TRIMMED VALUE BACK IN THE BOX
RETURN THE VALUE
END GET-VALUE
```

SHOW-RESULT (field name)

CALL THE CHECK FOR THAT FIELD
PUT WHAT IT RETURNS IN THE ERROR SPOT NEXT TO THE FIELD (FOUND BY ID)
IF THIS IS PASSWORD 1 AND PASSWORD 2 IS NOT EMPTY
    CALL SHOW-RESULT FOR PASSWORD 2
ENDIF
END SHOW-RESULT

SHOW-SLIDER-VALUE

DISPLAY THE SLIDER NUMBER AS "N out of 10" NEXT TO THE BAR
END SHOW-SLIDER-VALUE

Field Checks

EVERY CHECK

CHECK ONLY ITS OWN FIELD
RETURN THE ERROR MESSAGE IF SOMETHING IS WRONG
RETURN NOTHING IF THE FIELD IS GOOD
LEAVE THE ERROR SPOT TO SHOW-RESULT

CHECK-FIRST-NAME

IF FIRST NAME IS EMPTY
    RETURN "First name is required"
ELSE IF IT HAS ANYTHING BUT LETTERS APOSTROPHES AND DASHES OR IS OVER 30 CHARS
    RETURN "Letters, apostrophes and dashes only"
ENDIF
END CHECK-FIRST-NAME

CHECK-MIDDLE-INIT

IF MIDDLE INITIAL IS NOT EMPTY AND IS NOT ONE LETTER
    RETURN "Middle initial has to be one letter"
ENDIF
END CHECK-MIDDLE-INIT

CHECK-LAST-NAME

IF LAST NAME IS EMPTY

```
        RETURN "Last name is required"
ELSE IF IT HAS ANYTHING BUT LETTERS APOSTROPHES DASHES OR NUMBERS 2 TO 5
        RETURN "Letters, apostrophes, dashes and numbers 2 to 5 only"
ENDIF
END CHECK-LAST-NAME
```

CHECK-DOB

```
IF DATE OF BIRTH IS EMPTY
        RETURN "Date of birth is required"
ELSE IF IT IS NOT MM/DD/YYYY
        RETURN "Please use MM/DD/YYYY"
ELSE
        SPLIT IT INTO MONTH DAY AND YEAR
        IF THAT DAY DOES NOT EXIST (LIKE 02/30)
                RETURN "That date doesn't exist"
        ENDIF
        GET TODAYS DATE (SAME DATE SHOWN IN THE BANNER)
        SET OLDEST DATE TO TODAY MINUS 120 YEARS
        IF DATE OF BIRTH IS AFTER TODAY
                RETURN "Can't be in the future"
        ELSE IF DATE OF BIRTH IS BEFORE THE OLDEST DATE
                RETURN "Can't be more than 120 years ago"
        ENDIF
ENDIF
END CHECK-DOB
```

CHECK-SSN

```
IF SSN IS EMPTY
        RETURN "Social Security number is required"
ELSE IF IT IS NOT 9 DIGITS (DASHES OK)
        RETURN "SSN needs 9 digits"
ENDIF
END CHECK-SSN
```

CHECK-ADDRESS (address 1, address 2 and city)

```
IF ADDRESS 1 OR CITY IS EMPTY
        RETURN "Required"
ELSE IF ADDRESS 1 OR CITY IS LESS THAN 2 OR MORE THAN 30 CHARS
        RETURN "Needs 2 to 30 characters"
ENDIF
IF ADDRESS 2 IS NOT EMPTY AND IS LESS THAN 2 OR MORE THAN 30 CHARS
```

```
        RETURN "Needs 2 to 30 characters or leave it blank"
ENDIF
END CHECK-ADDRESS
```

CHECK-STATE

```
IF STATE IS STILL "Select"
        RETURN "Please pick a state"
ENDIF
END CHECK-STATE
```

CHECK-ZIP

```
IF ZIP IS EMPTY
        RETURN "Zip code is required"
ELSE IF IT IS NOT 5 DIGITS OR 5 DIGITS DASH 4 DIGITS
        RETURN "Zip code needs 5 digits"
ELSE
        CUT IT TO THE FIRST 5 DIGITS
        REDISPLAY THE SHORTER ZIP IN THE BOX
ENDIF
END CHECK-ZIP
```

CHECK-PHONE

```
IF PHONE IS EMPTY
        RETURN "Phone number is required"
ELSE IF IT IS NOT 10 DIGITS
        RETURN "Phone number needs 10 digits"
ELSE
        TAKE OUT EVERYTHING THAT IS NOT A DIGIT
        REDISPLAY IT AS 000-000-0000
ENDIF
END CHECK-PHONE
```

CHECK-EMAIL

```
IF EMAIL IS EMPTY
        RETURN "Email address is required"
ELSE IF IT IS NOT name@domain.tld
        RETURN "Email needs to look like name@domain.com"
ENDIF
END CHECK-EMAIL
```

CHECK-USER-ID

```
IF USER ID IS EMPTY
        RETURN "User ID is required"
ELSE IF IT HAS A SPACE
        RETURN "No spaces"
ELSE IF IT STARTS WITH A NUMBER
        RETURN "Can't start with a number"
ELSE IF IT IS NOT 5 TO 30 LETTERS NUMBERS UNDERSCORES OR DASHES
        RETURN "5 to 30 letters, numbers, underscores or dashes"
ELSE
        CHANGE IT TO LOWERCASE
        REDISPLAY THE LOWERCASE USER ID IN THE BOX
ENDIF
END CHECK-USER-ID
```

CHECK-PASSWORD-1 (runs as the user types)

```
IF PASSWORD IS EMPTY
        RETURN "Password is required"
ENDIF
START A LIST OF WHAT IS MISSING
IF LESS THAN 8 OR MORE THAN 30 CHARS ADD "8 to 30 characters"
IF NO UPPERCASE LETTER ADD "an uppercase letter"
IF NO LOWERCASE LETTER ADD "a lowercase letter"
IF NO NUMBER ADD "a number"
IF NO SPECIAL CHARACTER ADD "a special character"
IF IT HAS A DOUBLE QUOTE
        RETURN "No double quotes"
ELSE IF THE LIST IS NOT EMPTY
        RETURN "Still needs" AND THE LIST
ELSE IF IT HAS THE USER ID IN IT
        RETURN "Can't include your user ID"
ELSE IF IT HAS THE FIRST OR LAST NAME IN IT
        RETURN "Can't include your name"
ENDIF
END CHECK-PASSWORD-1
```

CHECK-PASSWORD-2 (runs as the user types)

```
IF PASSWORD 2 IS EMPTY
        RETURN "Please type your password again"
ELSE IF PASSWORD 2 DOES NOT EQUAL PASSWORD 1
        RETURN "Passwords do not match"
```

```
ENDIF
END CHECK-PASSWORD-2
```

CHECK-RADIO (gender, vaccinations, insurance)

```
IF NO BUTTON IN THE GROUP IS CHOSEN
        RETURN "Please answer this question"
ENDIF
END CHECK-RADIO
```

CHECK-SYMPTOMS

```
IF THE TEXT AREA HAS A DOUBLE QUOTE
        RETURN "Please leave out double quotes"
ENDIF
END CHECK-SYMPTOMS
```

Review

REVIEW-FORM

```
CALL EVERY CHECK AND KEEP THE RESULTS
SHOW EACH RESULT IN THE ERROR SPOT NEXT TO ITS FIELD
DISPLAY HEADING "PLEASE REVIEW YOUR INFORMATION"
LOOP THROUGH EVERY LINE OF THE REVIEW UNTIL NONE LEFT
        DISPLAY THE FIELD LABEL FROM THE PAGE (NOT THE VARIABLE NAME)
        CHECK THE KIND OF DATA
                CASE OF NAME
                        DISPLAY FIRST MI AND LAST TOGETHER
                END CASE
                CASE OF ADDRESS
                        DISPLAY ADDRESS 1, ADDRESS 2 IF THERE IS ONE, CITY STATE
AND ZIP
                END CASE
                CASE OF SSN
                        DISPLAY ONLY THE LAST 4 DIGITS
                END CASE
                CASE OF A RADIO BUTTON
                        DISPLAY THE WORDING OF THE BUTTON THAT WAS CHOSEN
                END CASE
                CASE OF A CHECKBOX
                        IF THIS ITEM WAS CHECKED
                                DISPLAY "Y"
```

```
                    ELSE
                        DISPLAY "N"
                    ENDIF
                END CASE
                CASE OF THE SLIDER
                    DISPLAY "N out of 10"
                END CASE
                CASE OF THE PASSWORD
                    DISPLAY STARS ONLY
                END CASE
                CASE OF BUTTONS
                    DO NOTHING
                END CASE
                DEFAULT CASE
                    DISPLAY VALUE
                END CASE
        IF THE CHECK FOR THIS LINE PASSED
                DISPLAY "pass"
        ELSE
                DISPLAY "ERROR" AND THE MESSAGE
        ENDIF
END LOOP
COUNT THE ERRORS
IF NO ERRORS
        DISPLAY "Everything looks good. Click Submit"
ELSE
        DISPLAY "Please fix the errors and click Review again"
ENDIF
RETURN THE NUMBER OF ERRORS
END REVIEW-FORM
```